

S1.2: Continuous Random Variables

1. Probability Density Function (PDF)

A continuous random variable X has a **probability density function** $f(x)$ satisfying

- $f(x) \geq 0$ for all x ,
- $\int_{-\infty}^{\infty} f(x) dx = 1$,
- $P(a \leq X \leq b) = \int_a^b f(x) dx$.

For continuous X , $P(X = x) = 0$ exactly — only intervals have non-zero probability.

X has pdf $f(x) = kx$ for $0 \leq x \leq 2$, 0 otherwise. Find k .

$$\int_0^2 kx dx = 2k = 1 \Rightarrow k = 1/2.$$

2. Cumulative Distribution Function (CDF)

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

Properties: non-decreasing, $F(-\infty) = 0$, $F(\infty) = 1$, and $f = F'$ (where differentiable).

Piecewise PDFs and CDFs

If $f(x)$ is given piecewise, compute $F(x)$ in each piece, matching values at the break points to keep F continuous.

For $f(x) = \frac{x}{2}$ on $[0, 2]$, find $F(x)$.

$F(x) = \int_0^x \frac{t}{2} dt = \frac{x^2}{4}$ for $0 \leq x \leq 2$. $F(x) = 0$ below 0, $F(x) = 1$ above 2.

3. Mean, Variance, Mode, Median

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

$$\text{Var}(X) = E(X^2) - \mu^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

Median m

Defined by $F(m) = \frac{1}{2}$. Solve the equation involving the CDF.

Mode

Value of x where $f(x)$ is maximum (stationary point or endpoint).

For $f(x) = \frac{x}{2}$ on $[0, 2]$:

$$\mu = \int_0^2 x \cdot \frac{x}{2} dx = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}.$$

$$E(X^2) = \int_0^2 x^2 \cdot \frac{x}{2} dx = \frac{1}{2} \cdot 4 = 2. \text{Var}(X) = 2 - \frac{16}{9} = \frac{2}{9}.$$

$$\text{Median: } F(m) = \frac{m^2}{4} = \frac{1}{2} \Rightarrow m = \sqrt{2}.$$

Mode: f is increasing on $[0, 2]$, so mode = 2 (endpoint).

4. Linear Transformations

$$E(aX + b) = aE(X) + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

CDF of a Transformed Variable

If $Y = aX + b$ with $a > 0$, then

$$F_Y(y) = P(Y \leq y) = P(X \leq (y - b)/a) = F_X\left(\frac{y - b}{a}\right).$$

Differentiate to get the pdf of Y .

5. Rectangular (Uniform) Distribution

$X \sim U(a, b)$ has pdf $f(x) = \frac{1}{b - a}$ for $a \leq x \leq b$.

- $E(X) = \frac{a + b}{2}$
- $\text{Var}(X) = \frac{(b - a)^2}{12}$
- $F(x) = \frac{x - a}{b - a}$ on the support.

6. Function of a Continuous Random Variable

To find the distribution of $Y = g(X)$:

1. Write $F_Y(y) = P(g(X) \leq y)$.
2. Manipulate the inequality to get it in terms of X .
3. Express in terms of F_X .
4. Differentiate to get $f_Y(y)$.

$X \sim U(0, 1)$. Find the pdf of $Y = X^2$.

$$F_Y(y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = \sqrt{y} \text{ for } 0 \leq y \leq 1.$$

$$\text{Differentiate: } f_Y(y) = \frac{1}{2\sqrt{y}} \text{ for } 0 < y \leq 1.$$

7. Exam Style Strategy

- Always check the pdf integrates to 1 before anything else.
- For medians of piecewise distributions, figure out which piece the median lies in first (which interval crosses $F = 0.5$).
- Express answers exactly when possible; decimal approximations can mask errors.
- When computing Var , use $E(X^2) - \mu^2$, not the definitional integral of $(x - \mu)^2$.